

Display Types -- Guided Notes

Unit tp-2.4 | CompTIA Network+ N10-009 Obj. FC0-U71 2.4 | N10-008 FC0-U71 2.4

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

LCD (Liquid Crystal Display) monitors use a _____ to illuminate a liquid crystal panel. LEDs are used as the backlight in most modern LCDs. The liquid crystals themselves _____ (produce/block and filter) light to create the image.

Question 2

OLED (Organic Light Emitting Diode) displays differ from LCDs because each pixel _____ its own light -- there is no separate backlight. This allows OLED to produce perfect _____ (true 0 luminance) by simply turning pixels off. The downside is potential _____ from prolonged static images.

Question 3

Resolution is measured in _____ x _____ (horizontal x vertical). Common resolutions include 1920x1080 (_____ or Full HD), 2560x1440 (_____), and 3840x2160 (_____ or 4K UHD). Higher resolution displays show _____ detail but require more GPU power.

Question 4

Refresh rate is measured in _____ (Hz) and indicates how many times per second the display updates the image. A 60 Hz monitor updates _____ times per second. A 144 Hz gaming monitor updates _____ times per second, making fast motion appear _____ and more responsive.

Question 5

Response time is measured in _____ (ms) and indicates how quickly a pixel can change from one color to another. Lower response time reduces _____, a blurring effect seen behind fast-moving objects. For competitive gaming, a response time of _____ ms or less is generally preferred.

Question 6

Three major display connector types are _____ (older analog standard), _____ (digital, carries audio and video, widely used), and _____ (high-speed digital, supports high refresh rates and daisy-chaining). Modern GPUs and monitors have moved away from _____ because it is an analog signal.

Question 7

Brightness is measured in _____ (nits). A typical office monitor is around 250-350 nits. HDR (High Dynamic Range) content requires _____ peak brightness to show the difference between dark and bright areas. Viewing angle determines how far off-center a user can sit before _____ shifts.

Question 8 -- Real World Application

REAL WORLD: A graphic designer is buying a new monitor for photo and video editing work. They are choosing between a high-refresh-rate gaming monitor (165 Hz, TN panel) and an IPS panel color-accuracy monitor (60 Hz, wide color gamut). Which should they choose and why? What is the most important spec for their use case?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-4-display-types/tp-2-4-display-types-answer-key.pdf
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